

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET E — CASE STUDY & PRACTICAL EXPERIMENTS

CHAPTER 06: HOW FORCES AFFECT MOTION & LAWS OF MOTION

TIME ALLOWED: 45 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-E-06

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET E - CASE STUDY & PRACTICAL EXPERIMENTS):

1. The question paper comprises **5 In-Depth Practical Case Studies** with **5 MCQs each (Total: 25 Questions, 1 Mark each)**.
2. Read each laboratory experiment, data table, and passage carefully before answering the corresponding questions.
3. Each question has 4 options with a single correct answer.
4. All questions are compulsory. There is no negative marking.
5. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

CASE STUDY 1: VERIFICATION OF NEWTON'S SECOND LAW ($F = MA$) WITH DYNAMIC TROLLEY

In a physics mechanics practical, students verified Newton's second law using a low-friction dynamic trolley of mass $M = 0.80 \text{ kg}$ on a horizontal runway.

- A light inextensible string tied to the trolley passed over a smooth pulley at the edge of the bench, carrying a hanging accelerating mass m .
 - The total mass of the accelerating system was kept strictly constant at $(M + m) = 1.00 \text{ kg}$ by transferring slotted masses between the trolley and the hanging holder.
 - Acceleration (a) was measured using ticker-timer tapes for four different accelerating weights ($F = mg$, where $g = 9.8 \text{ m/s}^2$):
- Trial 1: $F_1 = 0.49 \text{ N} \rightarrow a_1 = 0.49 \text{ m/s}^2$
 - Trial 2: $F_2 = 0.98 \text{ N} \rightarrow a_2 = 0.98 \text{ m/s}^2$
 - Trial 3: $F_3 = 1.47 \text{ N} \rightarrow a_3 = 1.47 \text{ m/s}^2$
 - Trial 4: $F_4 = 1.96 \text{ N} \rightarrow a_4 = 1.96 \text{ m/s}^2$

Q.1 What mathematical relationship between applied accelerating force F and acceleration a is validated by the experimental data?

- | | |
|---|---|
| <p>(A) Acceleration is inversely proportional to force ($a \propto 1/F$)</p> <p>(C) Acceleration is independent of applied force</p> | <p>(B) Acceleration is directly proportional to applied force ($a \propto F$) for constant system mass</p> <p>(D) Acceleration is proportional to the square of force ($a \propto F^2$)</p> |
|---|---|

Q.2 What is the slope of the graph plotted between applied force F (y-axis) and acceleration a (x-axis)?

- | | |
|---|--|
| <p>(A) 0.50 kg</p> <p>(C) 2.00 kg</p> | <p>(B) 1.00 kg</p> <p>(D) 9.8 kg</p> |
|---|--|

Q.3 Why was it crucial to transfer slotted masses from the trolley to the hanging hanger rather than simply adding new masses to the hanger?

- (A) To prevent the string from snapping
(B) To keep the total accelerated mass of the system $(M + m)$ strictly constant while systematically varying the accelerating force
(C) To reduce air resistance
(D) To make the trolley roll faster

Q.4 How did the students compensate for resistive friction along the runway before collecting data?

- (A) By pouring water onto the track
(B) By slightly tilting the runway so the component of gravity down the slope $(mg \sin \theta)$ exactly cancels friction
(C) By pushing the trolley continuously by hand
(D) By using rough sandpaper on the wheels

Q.5 If the accelerating force is held constant at 1.0 N while the total mass is doubled from 1.0 kg to 2.0 kg , what will the new acceleration be?

- (A) 0.50 m/s^2
(B) 1.00 m/s^2
(C) 2.00 m/s^2
(D) 0.25 m/s^2

CASE STUDY 2: IMPULSE-MOMENTUM THEOREM & AUTOMOTIVE CRASH ATTENUATION

Automotive safety engineers conducted crash-test simulations using sensor-equipped crash test dummies to assess passenger survival during head-on collisions.

- In the simulation, a car of mass 1200 kg travelling at $54 \text{ km/h} = 15 \text{ m/s}$ struck a rigid barrier and came to rest ($v=0$).
 - Unprotected Passenger (Dummy X):** Struck the rigid steering column directly, bringing head momentum to zero in a collision duration of $\Delta t_1 = 0.010 \text{ s}$.
 - Protected Passenger (Dummy Y):** Protected by a deployed airbag and front crumple zone, extending the head deceleration duration to $\Delta t_2 = 0.150 \text{ s}$.
- The head mass of both dummies was $m_{\text{head}} = 5.0 \text{ kg}$.

Q.6 What is the initial linear momentum of the dummy's head prior to impact?

- (A) $50 \text{ kg} \cdot \text{m/s}$
(B) $75 \text{ kg} \cdot \text{m/s}$
(C) $150 \text{ kg} \cdot \text{m/s}$
(D) $18000 \text{ kg} \cdot \text{m/s}$

Q.7 What is the magnitude of the impulse (Δp) experienced by the dummy's head during the collision?

- (A) $75 \text{ N} \cdot \text{s}$
(B) $150 \text{ N} \cdot \text{s}$
(C) $0 \text{ N} \cdot \text{s}$
(D) $500 \text{ N} \cdot \text{s}$

Q.8 What peak average impact force acted on the head of Dummy X (unprotected, $\Delta t_1 = 0.010 \text{ s}$)?

- (A) 750 N
(B) 7500 N
(C) 75 N
(D) 1500 N

Q.9 What peak average impact force acted on the head of Dummy Y (protected by airbag, $\Delta t_2 = 0.150 \text{ s}$)?

- (A) 500 N
(B) 7500 N
(C) 1500 N
(D) 75 N

Q.10 Which athletic technique in cricket directly applies this exact impulse-momentum principle?

- (A) A bowler running fast to bowl
(B) A fielder pulling his hands backwards while catching a fast cricket ball to prolong impact time and reduce hand force
(C) A batsman hitting a six
(D) An umpire raising his finger

CASE STUDY 3: VERIFICATION OF CONSERVATION OF LINEAR MOMENTUM ON AN AIR TRACK

Students verified the law of conservation of linear momentum using two gliders on a near-frictionless linear air track equipped with photogate timers.

- **Glider 1:** Mass $m_1 = 0.300 \text{ kg}$, travelling towards Glider 2 with velocity $u_1 = +0.80 \text{ m/s}$.
- **Glider 2:** Mass $m_2 = 0.500 \text{ kg}$, initially stationary at rest ($u_2 = 0 \text{ m/s}$).

The gliders were fitted with velcro bumpers so that upon colliding, they stuck together and moved as a single combined mass ($m_1 + m_2$) with common final velocity v .

Q.11 What is the total linear momentum of the two-glider system before the collision?

- (A) $0.150 \text{ kg} \cdot \text{m/s}$
(B) $0.240 \text{ kg} \cdot \text{m/s}$
(C) $0.400 \text{ kg} \cdot \text{m/s}$
(D) $0 \text{ kg} \cdot \text{m/s}$

Q.12 According to the law of conservation of momentum, what is the common final velocity v of the combined gliders after collision?

- (A) $+0.30 \text{ m/s}$
(B) $+0.48 \text{ m/s}$
(C) $+0.24 \text{ m/s}$
(D) $+0.80 \text{ m/s}$

Q.13 What was the total kinetic energy of the system before the collision?

- (A) 0.096 J
(B) 0.192 J
(C) 0.048 J
(D) 0.240 J

Q.14 What was the total kinetic energy of the combined gliders after the inelastic collision?

- (A) 0.096 J
(B) 0.036 J
(C) 0.072 J
(D) 0.018 J

Q.15 Why is total kinetic energy not conserved in this collision, even though linear momentum is conserved?

- (A) Because momentum was destroyed
(B) Because in an inelastic collision, some mechanical kinetic energy is converted into internal thermal energy and sound during deformation
(C) Because the air track lost pressure
(D) Because gravity increased

CASE STUDY 4: ACTION-REACTION FORCE VERIFICATION USING INTERCONNECTED SPRING BALANCES

To investigate Newton's third law of motion, students linked two identical spring balances, Balance A and Balance B, hook-to-hook.

- The ring of Balance A was anchored firmly to a rigid wall hook.
- A student held the ring of Balance B and pulled it horizontally to the right with varying forces.
- The force readings on both scales were recorded simultaneously:
 - Reading 1: Balance B pulled to 5.0 N $\rightarrow$ Balance A read 5.0 N

- Reading 2: Balance B pulled to 12.0 N $\rightarrow$ Balance A read 12.0 N
- Reading 3: Balance B pulled to 20.0 N $\rightarrow$ Balance A read 20.0 N

Q.16 What fundamental law of physics is directly validated by the identical readings on both spring balances?

- (A) Newton's First Law of Motion (B) Newton's Third Law of Motion (To every action there is an equal and opposite reaction)
- (C) Law of Conservation of Mass (D) Pascal's Principle

Q.17 If Action and Reaction forces are always equal in magnitude and opposite in direction, why do they not cancel each other to produce zero acceleration?

- (A) Because action force is always stronger than reaction force (B) Because Action and Reaction forces act simultaneously on two different, distinct physical bodies
- (C) Because reaction force occurs after a time delay (D) Because they act at an angle of 90°

Q.18 Which of the following real-world phenomena is an application of Newton's third law of motion?

- (A) A rocket lifting off by expelling high-velocity exhaust gases downward (B) Recoil of a rifle when a bullet is fired
- (C) A swimmer pushing water backward to propel himself forward (D) All of the above

Q.19 When a person of mass 60 kg stands on horizontal ground, what is the reaction force exerted by the ground on his feet ($g = 9.8\text{ m/s}^2$)?

- (A) 0 N (B) 588 N upward
- (C) 588 N downward (D) 60 N upward

Q.20 If the wall anchor breaks and both interconnected balances are pulled horizontally to the right accelerating across a frictionless floor, will their mutual contact readings still be equal?

- (A) No, the front balance will read zero (B) Yes, Newton's third law applies universally whether bodies are at rest or accelerating
- (C) The rear balance will read twice as much (D) Both balances will break

CASE STUDY 5: LIMITING STATIC FRICTION & KINETIC FRICTION ON AN INCLINED PLANE

Students placed a wooden block of mass $m = 0.50\text{ kg}$ on an adjustable wooden ramp to investigate friction.

- The angle of inclination (θ) of the ramp was gradually increased from 0° until the block just began to slide down spontaneously.
- The critical angle where sliding initiated was recorded as $\theta_s = 30^\circ$ (Angle of Repose).
- Once sliding, the ramp angle was adjusted to $\theta_k = 25^\circ$ so the block slid down with uniform constant velocity.
- Take $g = 9.8\text{ m/s}^2$, $\sin 30^\circ = 0.50$, $\cos 30^\circ = 0.866$, $\tan 30^\circ = 0.577$, $\tan 25^\circ = 0.466$.

Q.21 What is the coefficient of limiting static friction (μ_s) between the wooden block and the ramp surface?

- (A) 0.500 (B) 0.577
- (C) 0.866 (D) 0.466

Q.22 What is the coefficient of kinetic friction (μ_k) between the sliding block and the ramp?

- (A) 0.466 (B) 0.577
(C) 0.250 (D) 0.980

Q.23 Why is the coefficient of kinetic friction ($\mu_k = 0.466$) smaller than the coefficient of static friction ($\mu_s = 0.577$)?

- (A) Because mass decreases when moving (B) Because once an object is in motion, microscopic surface irregularities do not have sufficient time to interlock as deeply as when stationary
(C) Because gravity decreases during motion (D) Because air pushes the object

Q.24 What is the normal reaction force (N) exerted by the ramp on the 0.50 kg block at $\theta = 30^\circ$?

- (A) 4.90 N (B) 4.24 N
(C) 2.45 N (D) 9.80 N

Q.25 What net downward accelerating force acts on the block if the incline is raised to 45° (take $\sin 45^\circ = \cos 45^\circ = 0.707$ and $\mu_k = 0.466$)?

- (A) 1.85 N (B) 3.46 N
(C) 0.50 N (D) 4.90 N

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET E)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & PRACTICAL INVESTIGATION EXPLANATIONS (SET E)

Q. No.	Correct	Scientific Rationale & Experimental Data Analysis
Q.1	(B)	The ratio $\frac{F}{a} = \frac{0.49}{0.49} = \frac{0.98}{0.98} = 1.00 \text{ kg}$ is constant, proving $a \propto F$.
Q.2	(B)	Since $F = ma$, plotting F against a yields slope $m_{\text{total}} = \frac{\Delta F}{\Delta a} = 1.00 \text{ kg}$.
Q.3	(B)	To verify $a \propto F$ via fair testing, the total mass must be held constant. Transferring masses ensures $(M+m)$ remains invariant.
Q.4	(B)	Friction-compensating the runway involves tilting it just enough so the trolley rolls down at constant velocity when given a slight tap.
Q.5	(A)	$a = \frac{F}{m} = \frac{1.0 \text{ N}}{2.0 \text{ kg}} = 0.50 \text{ m/s}^2$. Acceleration is inversely proportional to mass.
Q.6	(B)	$p = m_{\text{head}} \times u = 5.0 \text{ kg} \times 15 \text{ m/s} = 75 \text{ kg} \cdot \text{m/s}$.
Q.7	(A)	Impulse $= \Delta p = p_f - p_i = 0 - 75 = 75 \text{ N} \cdot \text{s}$ (or $\text{kg} \cdot \text{m/s}$).
Q.8	(B)	$F_1 = \frac{\Delta p}{\Delta t_1} = \frac{75 \text{ N} \cdot \text{s}}{0.010 \text{ s}} = 7500 \text{ N}$ (sufficient to cause fatal skull fracture).
Q.9	(A)	$F_2 = \frac{\Delta p}{\Delta t_2} = \frac{75 \text{ N} \cdot \text{s}}{0.150 \text{ s}} = 500 \text{ N}$ (reduced by 15-fold, preventing fatal trauma).
Q.10	(B)	Drawing hands backward increases the time Δt over which the ball's momentum drops to zero, reducing the impact force $F = \Delta p / \Delta t$ on the hands.
Q.11	(B)	$p_{\text{total}} = m_1 u_1 + m_2 u_2 = (0.300 \times 0.80) + (0.500 \times 0) = 0.240 \text{ kg} \cdot \text{m/s}$.
Q.12	(A)	$p_f = (m_1 + m_2)V = (0.300 + 0.500)V = 0.800 V$. By conservation: $0.800 V = 0.240 \Rightarrow V = \frac{0.240}{0.800} = +0.30 \text{ m/s}$.
Q.13	(A)	$E_{k, \text{initial}} = \frac{1}{2} m_1 u_1^2 = \frac{1}{2} (0.300)(0.80)^2 = 0.150 \times 0.64 = 0.096 \text{ J}$.
Q.14	(B)	$E_{k, \text{final}} = \frac{1}{2} (m_1 + m_2) V^2 = \frac{1}{2} (0.800)(0.30)^2 = 0.400 \times 0.09 = 0.036 \text{ J}$.
Q.15	(B)	Total linear momentum is strictly conserved in all collisions without net external force. However, in inelastic collisions, kinetic energy is partially dissipated as heat, sound, and material deformation.
Q.16	(B)	The pull exerted by Balance B on Balance A (action) is matched by an equal and opposite pull exerted by Balance A on Balance B (reaction).
Q.17	(B)	Forces cancel only when acting on the same single body. Action acts on body 1 while reaction acts on body 2; hence, they cannot cancel each other.
Q.18	(D)	Rocket thrust, rifle recoil, and swimming propulsion all operate on mutual equal-and-opposite action-reaction pairs between interacting bodies.
Q.19	(B)	The person's feet exert a downward normal contact force of $mg = 60 \times 9.8 = 588 \text{ N}$ on the ground. By Newton's third law, the ground exerts an equal upward normal reaction of 588 N.
Q.20	(B)	Newton's third law is an invariant law of nature; interaction forces between two bodies are instantaneously equal and opposite at every instant, accelerating or at rest.
Q.21	(B)	At the angle of repose, component of gravity equals limiting static friction: $mg \sin \theta_s = \mu_s mg \cos \theta_s \Rightarrow \mu_s = \tan \theta_s = \tan 30^\circ \approx 0.577$.
Q.22	(A)	When sliding at constant velocity, net force is zero: $\mu_k = \tan \theta_k = \tan 25^\circ \approx 0.466$.
Q.23	(B)	Static friction requires breaking cold welds between microscopic surface asperities; once in motion, asperities skip over each other, reducing kinetic friction.
Q.24	(B)	$N = mg \cos 30^\circ = (0.50 \times 9.8) \times 0.866 = 4.90 \times 0.866 \approx 4.24 \text{ N}$.

Q. No.	Correct	Scientific Rationale & Experimental Data Analysis
Q.25	(A)	$F_{\text{net}} = mg \sin 45^\circ - \mu_k mg \cos 45^\circ = mg \sin 45^\circ (1 - \mu_k) = (4.90 \times 0.707) \times (1 - 0.466) = 3.464 \times 0.534 \approx 1.85 \text{ N}$